AWS Cloud Practitioner

Course Objective

Course Objectives:

- To understand of cloud computing concepts using AWS console
- To design and implement effective networking solutions and implement robust security measures to protect AWS resources and data.
- To analyze proficiency in managing storage solution and database on the AWS platform.
- To evaluate pricing models and support options to make informed decisions on resource allocation and budgeting.

Module -1

- **Cloud Computing Terminology** – Cloud Computing Foundation, Essential Characteristics, Architectural Overview, Components of cloud computing, Cloud Delivery Model, Deployment Model, Clous Service Model – SaaS, IaaS, PaaS.

- **AWS Console** – AWS Cloud History, AWS Global Infrastructure – AWS Regions, Availability Zones, Data Centers, and Edge Locations, and Tour of the AWS Console.

History of Cloud Computing

One of the most questions comes in our mind "when and where was the first premise of cloud computing witnessed?"

- **1950-1960:**

  The concept of time sharing was introduced to allow multiple users to gain shared access to data and CPU time. This was done to lower the cost of maintenance of large-sized computers.

- **1960-1970:**

  ARPANET (Advanced Research Projects Agency Network), the basis of internet was introduced. The basis of staying connected and accessing data from any location was established.

- **1970-1980:**

  IBM introduced the Virtual Machine OS, and this enabled users to have multiple virtual machines on a single physical node.

History of Cloud Computing(Contd..

- **1980-1990:**

  The time period between1980 and 1990 witnessed the emergence of Internet Service Providers and Application Service Providers, which broke the realm of desktops and single user server to deliver hosted application services.

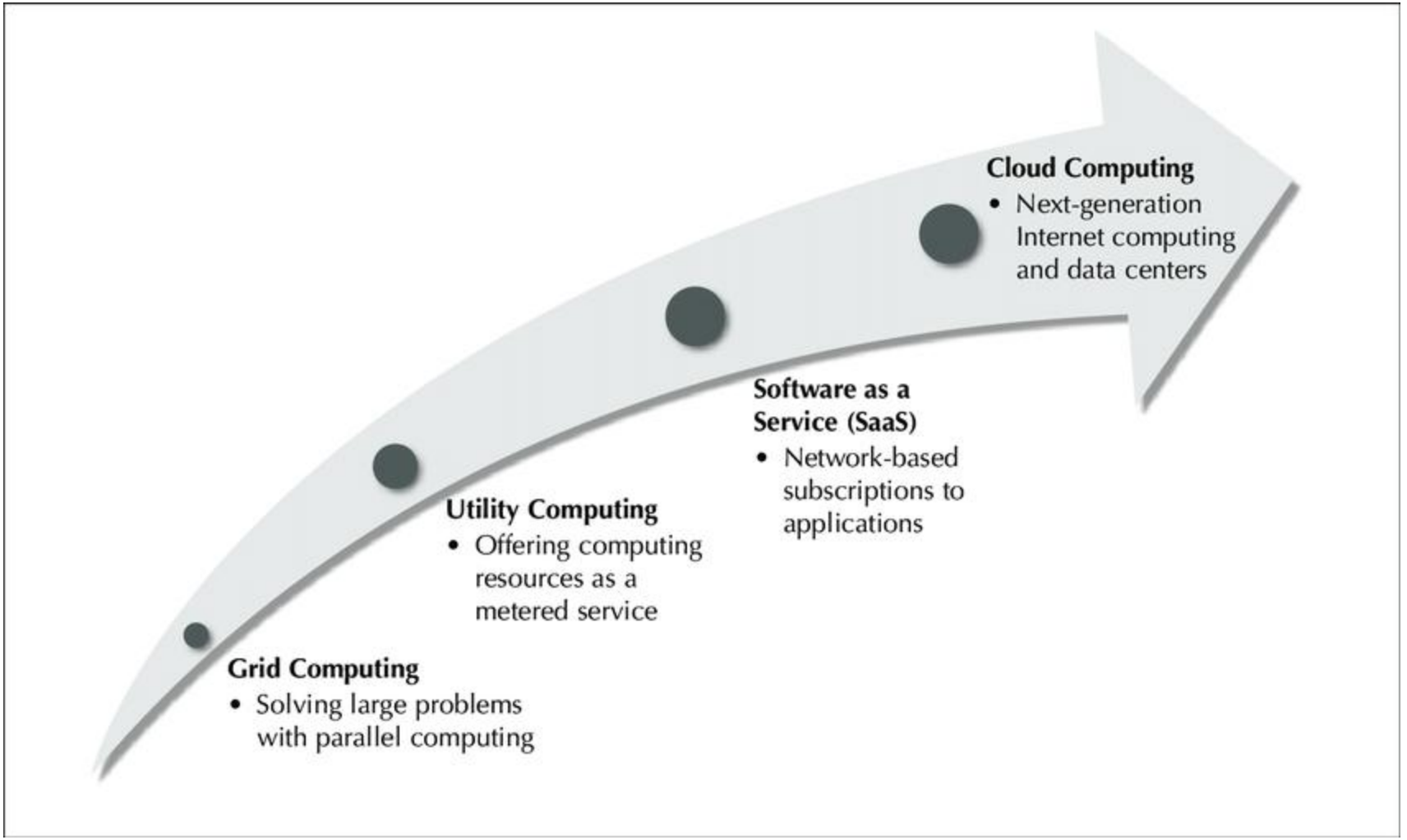
- **1990-2000:**

  The internet became more prevalent, and virtualized connections for PC-based systems grew to correct network traffic and bandwidth usage. Grid computing gained popularity along with object oriented programming and web services.

- **After 2000:**

  The 21st century witnessed tremendous growths of the cloud as more and more corporate solutions and services from players like IBM and Oracle came in the market.

Evolution of Cloud Computing(Contd..



Introduction to Cloud Computing

- The National Institute of Standards and Technology (NIST) defines Cloud Computing as:

"A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (for example networks, servers, storage, applications, etc) that can be rapidly provisioned and released with minimal management effort or service provider interaction.



Cloud Computing (Contd..



Fig. Cloud Computing

Figure shows the pictorial representation of Cloud Computing in detail. Critical data that takes different forms, such as emails, text document, videos, chats and presentations are stored, accessed and processes as needed by the the cloud user- which is a set of software and hardware devices (servers and storage) commissioned in a data centre, either in a single or multiple location(s).

Essential characteristics of Cloud Computing

- **On-demand self-service:** Cloud Computing empowers the consumer to unilaterally provision computing capabilities like server time and network storage with no need for human interaction with individual service providers.

- **Broad Network Access:** In cloud computing, the Internet allows any thin or thick client, such as laptop or a mobile phone, to access capabilities through standard mechanisms.

- **Resource pooling:** In the Cloud Environment, the provider's computing resources are pooled to serve multiple consumers through the multi-tenant model, with different resources being assigned and reassigned on a real-time basis according to consumer demand. The user loses control over the location of the data but may be able to specify a higher level of abstraction.

Essential characteristics of Cloud Computing(Contd..

- **Rapid Elasticity:** Capabilities can be both elastically and automatically provisioned and released, to scale rapidly with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.

- **Measured service:** Cloud Systems automatically monitor, control, optimise and report resource usage, thus providing transparency for both the provider and consumer of the utilised service.

Advantage of Cloud Computing

- Lower computer costs
- Improved performance
- Reduced software cost
- Instant software updates
- Unlimited storage capacity
- Increased data reliability
- Universal document access
- Latest version availability
- Device independence
- Reduced impact on environment
- Better agility

Disadvantage of Cloud Computing

- Requires a constant Internet connection
- Does not work well with low-speed connections
- Stored data can be lost
- Stored data might not be secure

Obstacles for Cloud Technology

- Data Security and privacy issues
- Adherence to regulatory and compliance measures
- Vendor lock-in
- Lack of performance and uptime
- Depending on outside agency
- Knowledge and integration

Cloud Vulnerabilities

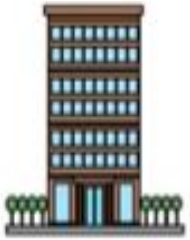
- Session riding
- Virtual Machine Escape
- Reliability and Availability
- Insecure cryptography
- Data Portability
- Vendor Locking
- Internet Dependency

Cloud Challenges

- Keeping up with security requirements
- Obtaining the right knowledge and expertise
- Choose the right vendor
- Data Interoperability
- Budgeting difficulties while moving to the cloud

On-premise vs Cloud Computing

ON-PREMISE



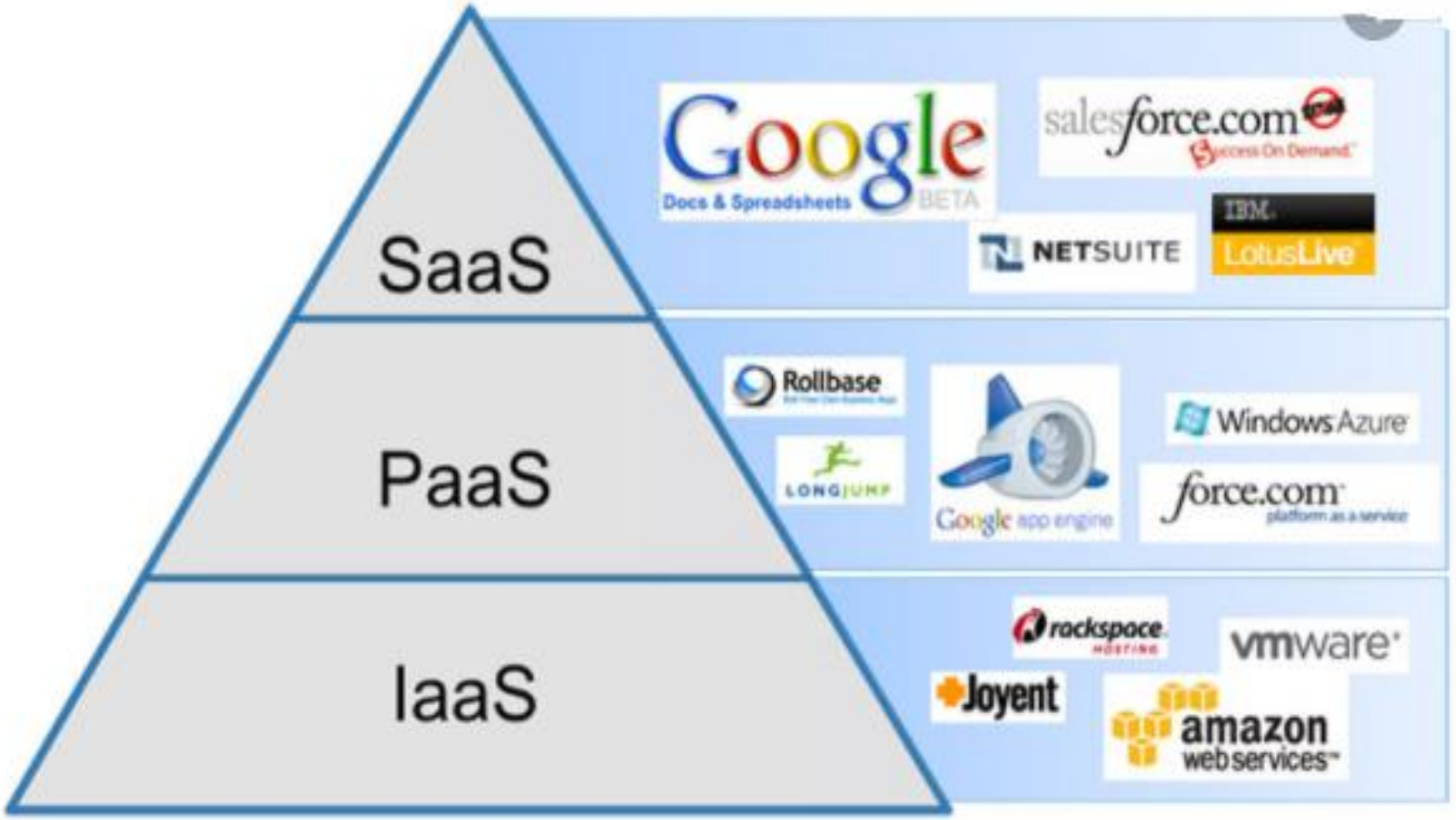


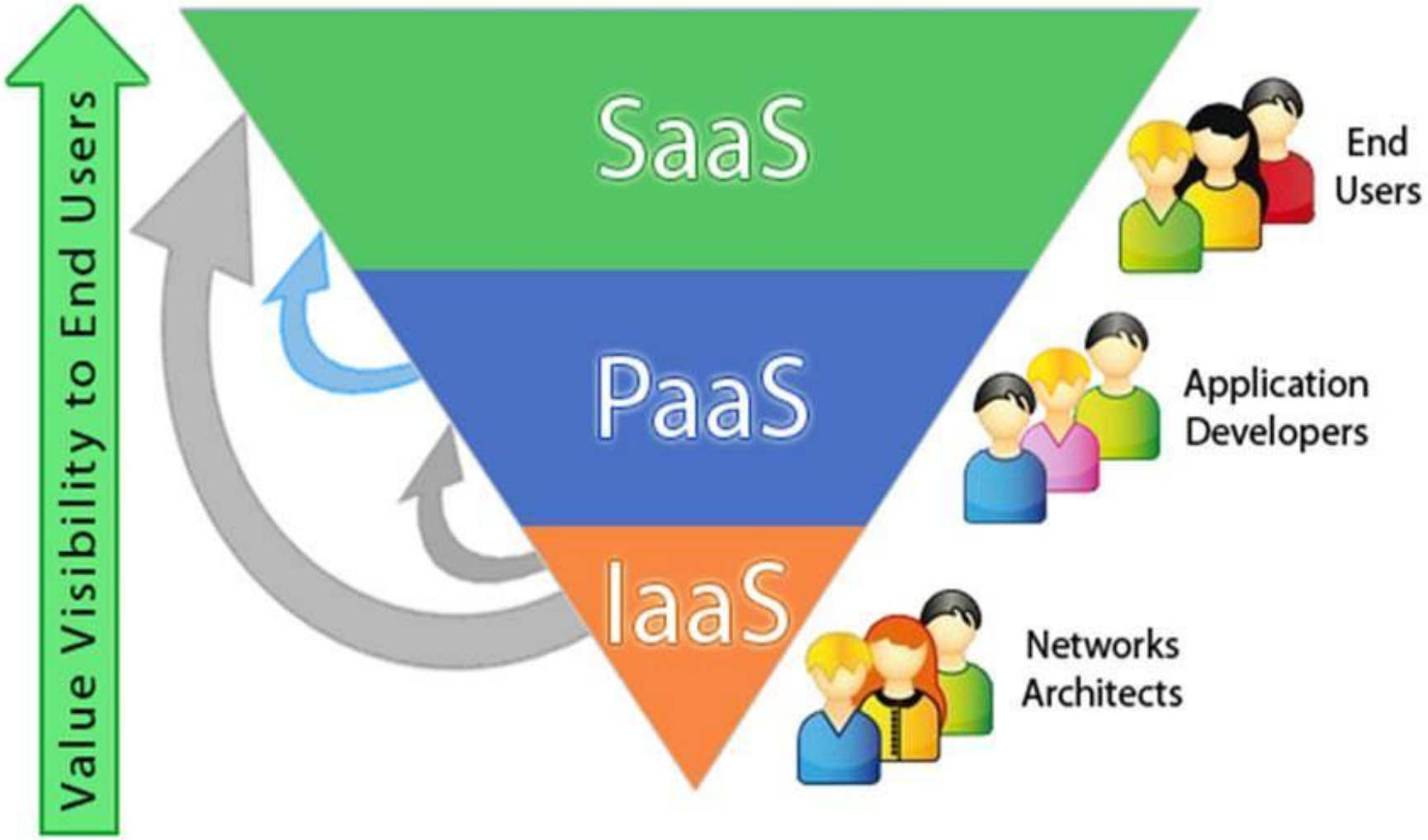
- Higher pay, less scalability
- Allot huge space for servers
- Appoint a team for hardware and software maintenance
- Poor data security
- Less chance of data recovery

- Pay for what you use

  Scale up= pay more
  Scale down= pay less
- No server space required
- No experts required for hardware and software maintenance
- Better data security
- Disaster recovery

Cloud Service/delivery Models




Value Visibility to End Users
SaaS
PaaS
IaaS
End Users
Application Developers
Networks Architects

Cloud Service Models (Contd..

- **Software as a Service (SaaS):** The ability to access the provider's applications running in the cloud environment is referred to as SaaS. Devices such as smart phones, laptops, desktops and tablets can be used to access the applications through a web browser or a program interface. The cloud user, however, cannot access or manage the cloud infrastructure that hosts the applications. This includes the servers, storage devices, network and individual application components.

Advantages of SaaS

- **Cost-Effective:** Pay only for what you use.
- **Reduced time:** Users can run most SaaS apps directly from their web browser without needing to download and install any software. This reduces the time spent in installation and configuration and can reduce the issues that can get in the way of the software deployment.
- **Accessibility:** We can Access app data from anywhere.
- **Automatic updates:** Rather than purchasing new software, customers rely on a SaaS provider to automatically perform the updates.
- **Scalability:** It allows the users to access the services and features on-demand.

Cloud Service Models (Contd..

- **Platform as a Service (PaaS):** PaaS assists a developer who writes custom application by provisioning the hardware, operating system, database and middleware. In the PaaS model, user do not have any control over the cloud infrastructure just like the SaaS model. However, they have the ability to monitor and manage the applications that they have deployed and the respective configuration settings for the application-hosting environment.

Advantages of PaaS:

- **Simple and convenient for users:** It provides much of the infrastructure and other IT services, which users can access anywhere via a web browser.
- **Cost-Effective:** It charges for the services provided on a per-use basis thus eliminating the expenses one may have for on-premises hardware and software.
- **Efficiently managing the lifecycle:** It is designed to support the complete web application lifecycle: building, testing, deploying, managing, and updating.
- **Efficiency:** It allows for higher-level programming with reduced complexity thus, the overall development of the application can be more effective.

Cloud Service Models (Contd..

- **Infrastructure as a Service (IaaS):** The user is allowed to provision processing, storage, networks and other fundamental computing resources where the consumer is able to deploy and run operating system and other applications. Though the cloud user can establish complete control over operating systems, storage and deployed applications, the underlying cloud infrastructure remains unperturbed.

Advantages of IaaS:

- **Cost-Effective:** Eliminates capital expense and reduces ongoing cost and IaaS customers pay on a per-user basis, typically by the hour, week, or month.
- **Website hosting:** Running websites using IaaS can be less expensive than traditional web hosting.
- **Security:** The IaaS Cloud Provider may provide better security than your existing software.
- **Maintenance:** There is no need to manage the underlying data center or the introduction of new releases of the development or underlying software. This is all handled by the IaaS Cloud Provider.

Difference between IAAS, PAAS and SAAS :

Basis Of	IAAS	PAAS	SAAS
Stands for	Infrastructure as a services.	Platform as a services.	Software as a services.
Uses	IAAS is used by network architects.	PAAS is used by developer.	SAAS is used by end user.
Access	IAAS give access to the resources like virtual machines and virtual storage.	PAAS give access to run time environment to deployment and development tools for application.	SAAS give access to the end user.
Model	It is service model that provide visualized computing resources over internet.	It is a cloud computing model that delivers tools that is used for development of application.	It is a service model in cloud computing that host software make available for client.

Difference between IAAS, PAAS and SAAS (Contd..:

Basis Of	IAAS	PAAS	SAAS
Technical understanding.	It required technical knowledge.	In which you required knowledge of subject to understand basic setup.	There is no requirement about technicalities company handle everything.
Popularity.	It is popular between developer and researchers.	It popular between developer who focus on the development of apps and scripts.	It is popular between consumer and company.such as file sharing, email and networking
Cloud services.	Amazon web services, sun, vcloud express.	Facebook, and google search engine.	M.S office web, Facebook and google apps.

Difference between IAAS, PAAS and SAAS (Contd..:

Basis Of	IAAS	PAAS	SAAS
Enterprise services.	AWS virtual private cloud.	Microsoft azure.	IBM cloud analysis.
Outsourced cloud services.	Salesforced.	Force.com, Gigaspaces.	AWS, terremark

Which cloud service is suitable for you?



IAAS

If your business needs a virtual machine, opt for Infrastructure as a Service



PAAS

If your company requires a platform for building software products, pick Platform as a Service

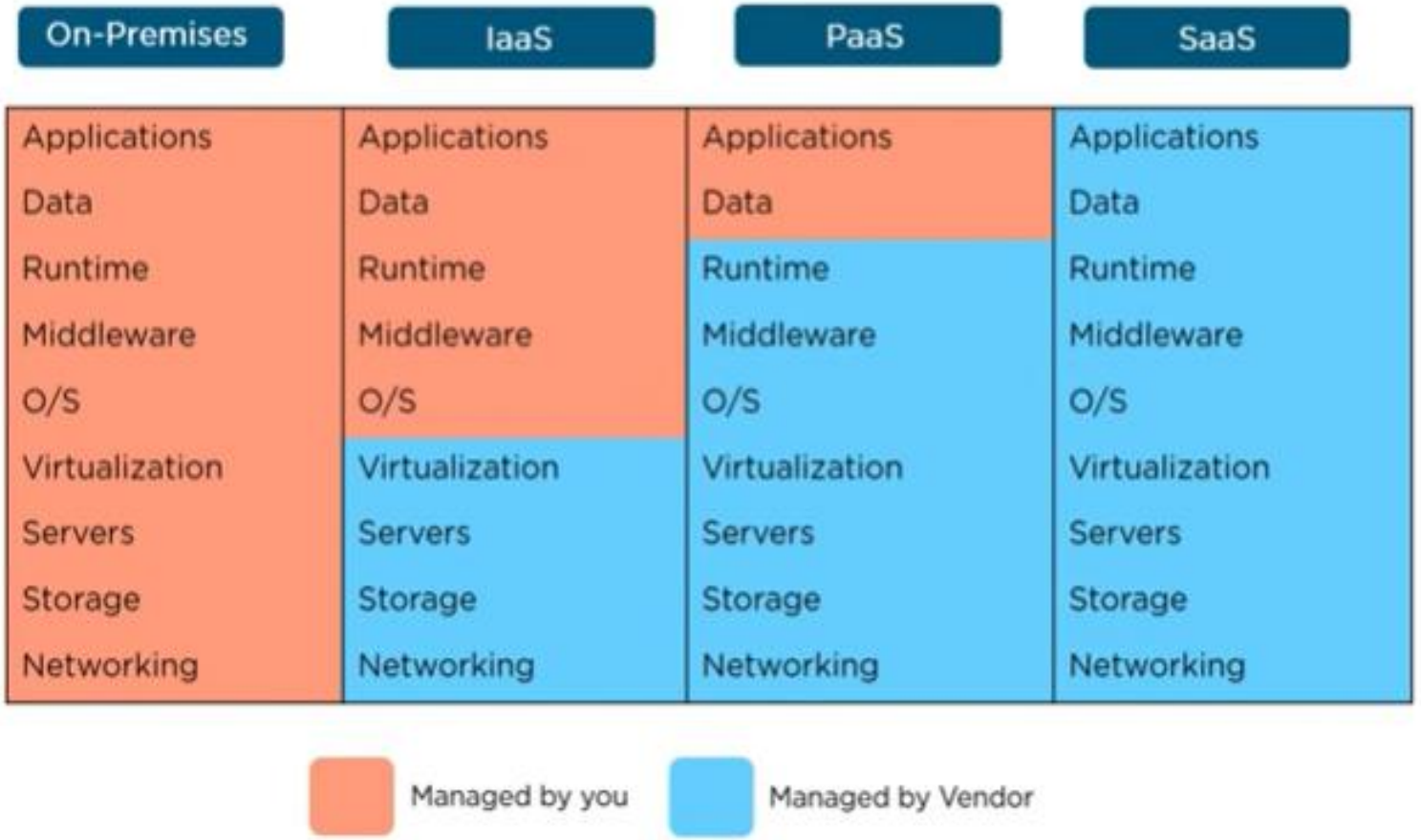


SAAS

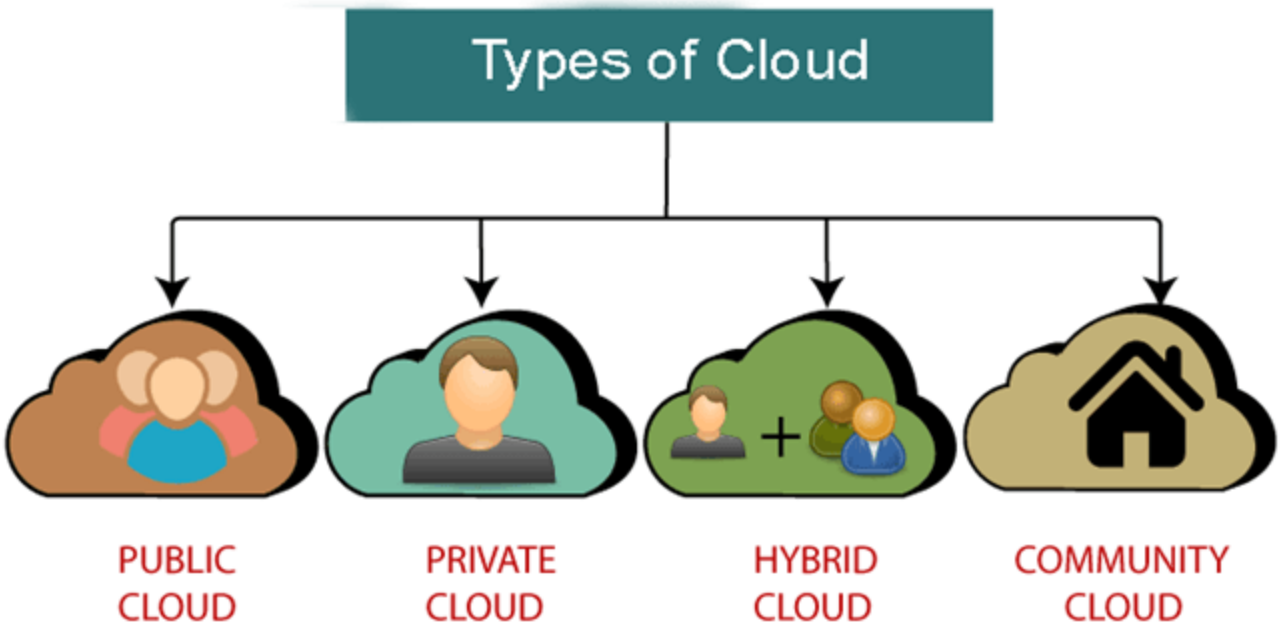
If your business doesn't want to maintain any IT equipment, then choose Software as a Service



Differences between IaaS, PaaS and SaaS



Types of Cloud/ Deployment Model



Public Cloud

- Public cloud is **open to all** to store and access information via the Internet using the pay-per-usage method.
- In public cloud, computing resources are managed and operated by the Cloud Service Provider (CSP).

Example:

Amazon elastic compute cloud (EC2),

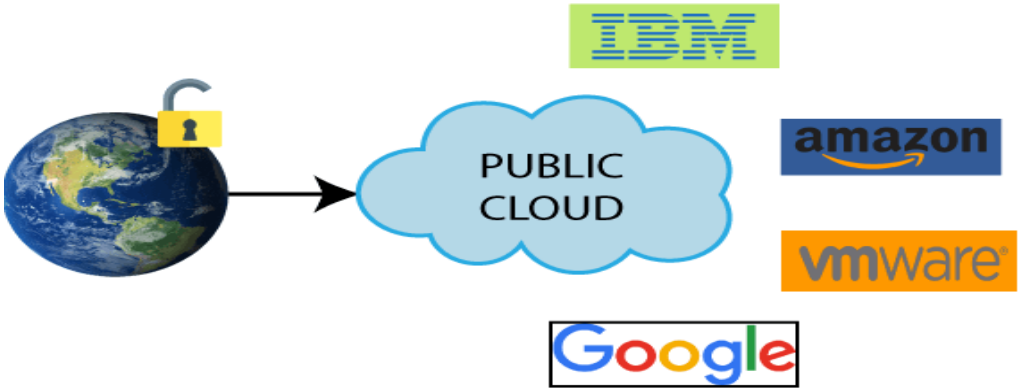
IBM Smart Cloud Enterprise,

Microsoft,

Google App Engine,

Windows Azure Services Platform.

Public Cloud



Advantage

1) Low Cost
Public cloud has a lower cost than private, or hybrid cloud, as it shares the same resources with a large number of consumers.

2) Location Independent
Public cloud is location independent because its services are offered through the internet.

3) Save Time
In Public cloud, the cloud service provider is responsible for the manage and maintain data centers in which data is stored, so the cloud user can save their time to establish connectivity, deploying new products, release product updates, configure, and assemble servers.

Public Cloud

4) Quickly and easily set up
Organizations can easily buy public cloud on the internet and deployed and configured it remotely through the cloud service provider within a few hours.

5) Business Agility
Public cloud provides an ability to elastically re-size computer resources based on the organization's requirements.

6) Scalability and reliability
Public cloud offers scalable (easy to add and remove) and reliable (24*7 available) services to the users at an affordable cost.

Disadvantage
1) Low Security
Public Cloud is less secure because resources are shared publicly.

2) Performance
In the public cloud, performance depends upon the speed of internet connectivity.

3) Less customizable
Public cloud is less customizable than the private cloud.

Private Cloud

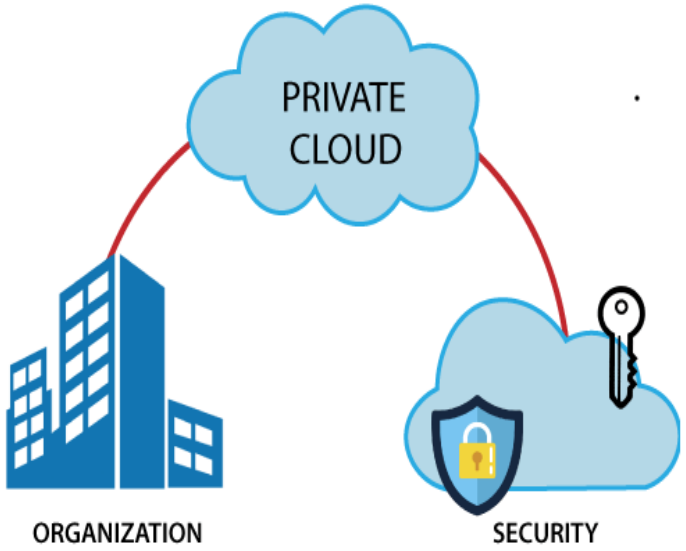
- Private cloud is also known as an **internal cloud** or **corporate cloud**.

- Private cloud provides computing services to a **private internal network (within the organization)** and **selected users** instead of the general public.

- Private cloud provides a **high level of security** and **privacy** to data through firewalls and internal hosting.

Examples:

- HP Data Centers
- Microsoft
- Ubuntu



Private Cloud

Advantages

1) More Control

Private clouds have more control over their resources and hardware than public clouds because it is only accessed by selected users.

2**) Security & privacy**

Security & privacy are one of the big advantages of cloud computing. Private cloud improved the security level as compared to the public cloud.

3) Improved performance

Private cloud offers better performance with improved speed and space capacity.

Private Cloud

Disadvantages

1) High cost
The cost is higher than a public cloud because set up and maintain hardware resources are costly.

2) Restricted area of operations
As we know, private cloud is accessible within the organization, so the area of operations is limited.

3) Limited scalability
Private clouds are scaled only within the capacity of internal hosted resources.

4) Skilled people
Skilled people are required to manage and operate cloud services.

Hybrid Cloud

- Hybrid cloud is a combination of **public and private** clouds.

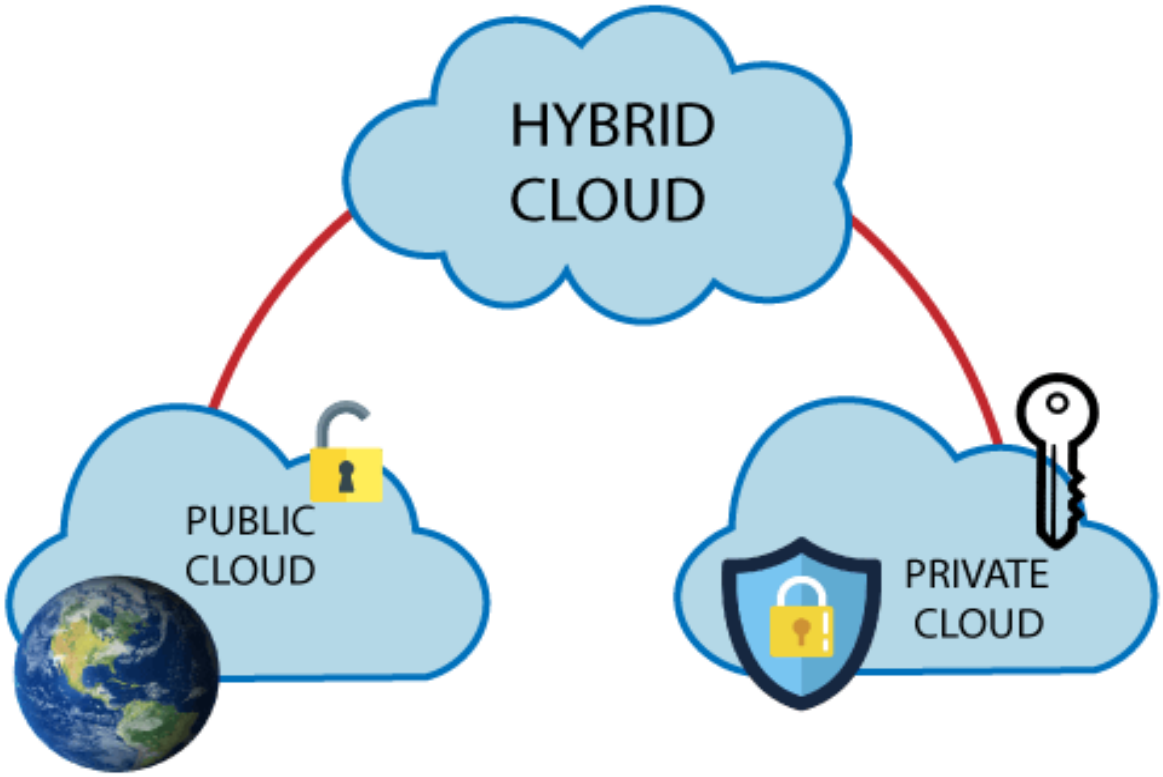
Hybrid cloud = public cloud + private cloud

- The main aim to combine these cloud (Public and Private) is to create a unified, automated, and well-managed computing environment.

- In the Hybrid cloud, **non-critical activities** are performed by the **public cloud** and **critical activities** are performed by the **private cloud**.

- Mainly, a hybrid cloud is used in finance, healthcare, and Universities.

- The best hybrid cloud provider companies are **Amazon, Microsoft, Google, Cisco,** and **NetApp**.


HYBRID
CLOUD
PUBLIC
CLOUD
PRIVATE
CLOUD

Hybrid Cloud

Advantages

1) Flexible and secure

It provides flexible resources because of the public cloud and secure resources because of the private cloud.

2) Cost effective

Hybrid cloud costs less than the private cloud. It helps organizations to save costs for both infrastructure and application support.

3) Cost effective

It offers the features of both the public as well as the private cloud. A hybrid cloud is capable of adapting to the demands that each company needs for space, memory, and system.

4) Security

Hybrid cloud is secure because critical activities are performed by the private cloud.

5) Risk Management

Hybrid cloud provides an excellent way for companies to manage the risk.

Hybrid Cloud

Disadvantages

1)Networking issues

In the Hybrid Cloud, networking becomes complex because of the private and the public cloud.

2) Infrastructure Compatibility

Infrastructure compatibility is the major issue in a hybrid cloud. With dual-levels of infrastructure, a private cloud controls the company, and a public cloud does not, so there is a possibility that they are running in separate stacks.

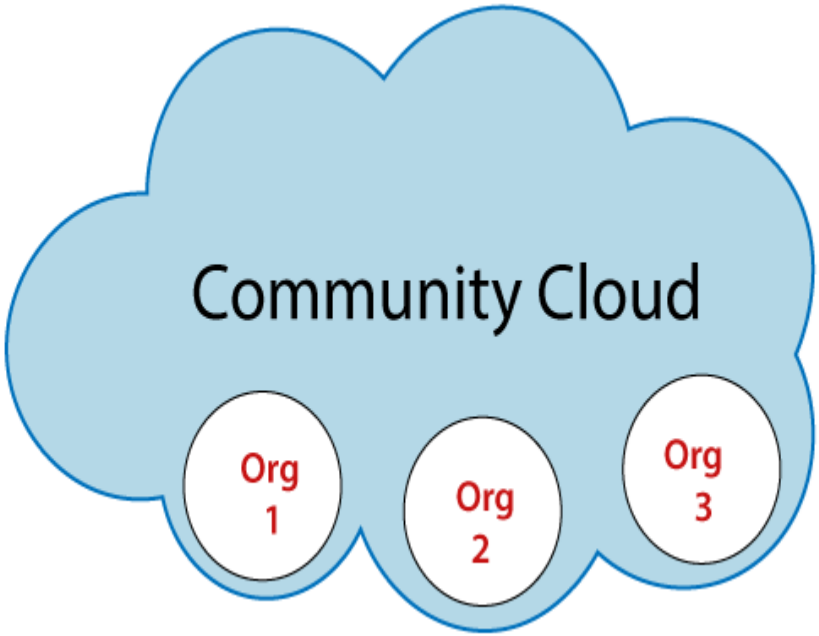
3) Reliability

The reliability of the services depends on cloud service providers.

Community Cloud

- It is a cloud infrastructure that allows systems and services to be accessible by a group of several organizations to share the information.

- It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.



Community Cloud

Advantages

- Cost effective

Community cloud is cost effective because the whole cloud is shared between several organizations or a community.

- Flexible and Scalable

The community cloud is flexible and scalable because it is compatible with every user. It allows the users to modify the documents as per their needs and requirement.

- Security

Community cloud is more secure than the public cloud but less secure than the private cloud.

- Sharing infrastructure

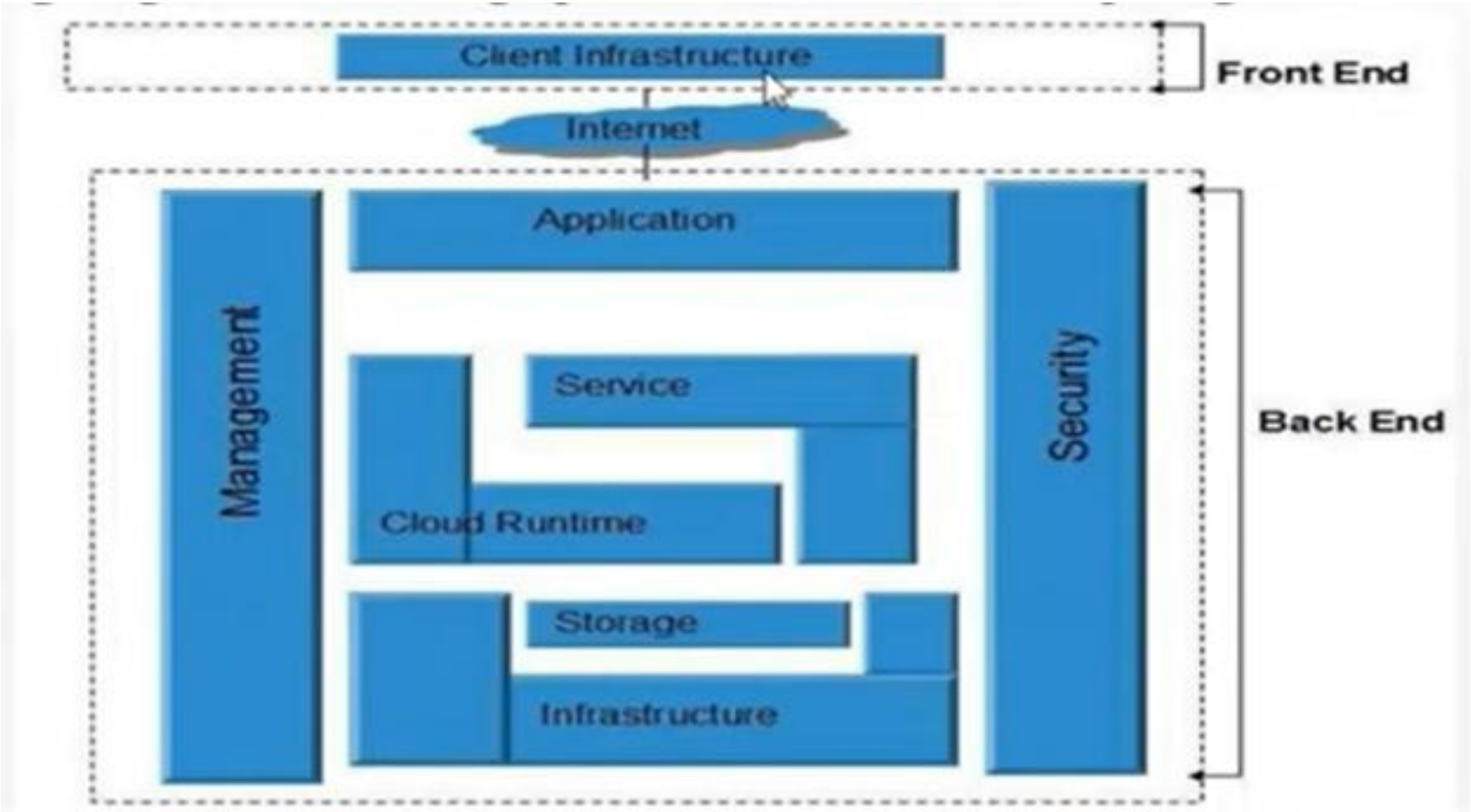
Community cloud allows us to share cloud resources, infrastructure, and other capabilities among various organizations.

Community Cloud

Disadvantages

- Community cloud is not a good choice for every organization.
- Slow adoption to data
- The fixed amount of data storage and bandwidth is shared among all community members.
- Community Cloud is costly than the public cloud.
- Sharing responsibilities among organizations is difficult.

Cloud Computing Architecture



Cloud Computing Architecture

1. **Front-End (Client-Side)**:

•**Description**: The interface that the user interacts with.

•**Components**:

- Web browsers or client devices (e.g., laptops, mobile apps).
- User interfaces or APIs for accessing cloud services.

•**Example**: AWS Management Console, mobile applications accessing cloud services.

Cloud Computing Architecture

2. Back-End (Cloud-Side):

•**Description**: The cloud infrastructure that powers applications and services.

•**Components**:

- **Compute**: Virtual machines (e.g., EC2), container services (e.g., Amazon ECS, Kubernetes).
- **Storage**: Object storage (e.g., S3), block storage (e.g., EBS), databases (e.g., RDS, DynamoDB).
- **Networking**: Virtual networks (e.g., VPC), load balancers, DNS services (e.g., Route 53).
- **Monitoring and Management Tools**: AWS CloudWatch, CloudTrail.
- **Security**: IAM, encryption, firewalls.

Cloud Computing Architecture

3. Cloud Resources:

•Hosted and managed by cloud providers, these include:

- **Data Centers**: Physical infrastructure.
- **Virtualization**: Hypervisors for resource abstraction.
- **Middleware**: Software that enables integration between applications.

Cloud Computing Architecture

4. Networking:

•The interconnection of cloud services, data centers, and client devices.

•**Key Elements**:

- Content Delivery Networks (e.g., CloudFront).
- Gateways, APIs, and firewalls.
- High-speed data transfer protocols.

Cloud Computing Architecture

5. Cloud Storage:

•**Types**:

- Object Storage: Scalable (e.g., S3).
- Block Storage: For databases and applications (e.g., EBS).
- File Storage: Shared storage (e.g., Amazon EFS).

6. Management and Monitoring:

•Tools to manage and optimize the cloud environment.

•Examples: AWS CloudFormation (automation), AWS Config (compliance).



(i). Brief about Amazon Web Services

Amazon Web Services (AWS) offers reliable and extensive cloud application and infrastructure services for all types of organisations. With over 10 years of experience in supporting enterprises with affordable cloud services AWS cloud has grown into a leader in cloud storage and management of various critical business processes.

Key things to know about AWS:

- AWS is the pioneer in the Cloud Computing race and has managed to retain its top position through the years.
- Recent price cuts have allured more startups as well as government agencies towards the AWS cloud.
- Amazon offers three types of computing instances: On-demand, reserved and spot. All three options come with different pricing structures.
- Amazon's cloud offering for Big data analytics is called Amazon Elastic Map Reduce.

(ii). Brief about Microsoft Azure



The enterprise-grade Cloud Computing platform from Microsoft is called Azure, and it supports a wide selection of tools, languages, frameworks, databases and devices. Statistics reveal that 57% of the Fortune 500 companies from various industry sectors entrust their data and applications with Azure. The easy-to-implement and low-cost hybrid cloud solution offered by Microsoft Azure is a good fit for organisations of all types and sizes.



(iii). Brief about Google Cloud Platform

The Google Cloud Platform allows users to build applications and store data on Google's highly reliable and scalable infrastructure. The innovation power of Google is world renowned, and by leveraging its cloud platform, developers can gain quicker access to its technological innovations .

Key things to know about Google Cloud Platform: .

- Google's network can be defined by its thousands of miles of fibre optic cables connected through cutting-edge Andromeda networking. The consistency and speed of such a network is unparalleled.
- Data is 100% secure through backup storage across multiple locations.
- The per-minute billing system allows users to handle traffic spikes with the most cost-effective solutions.
- The Big Query and Google Cloud Dataflow tools offer full support to big data solutions.

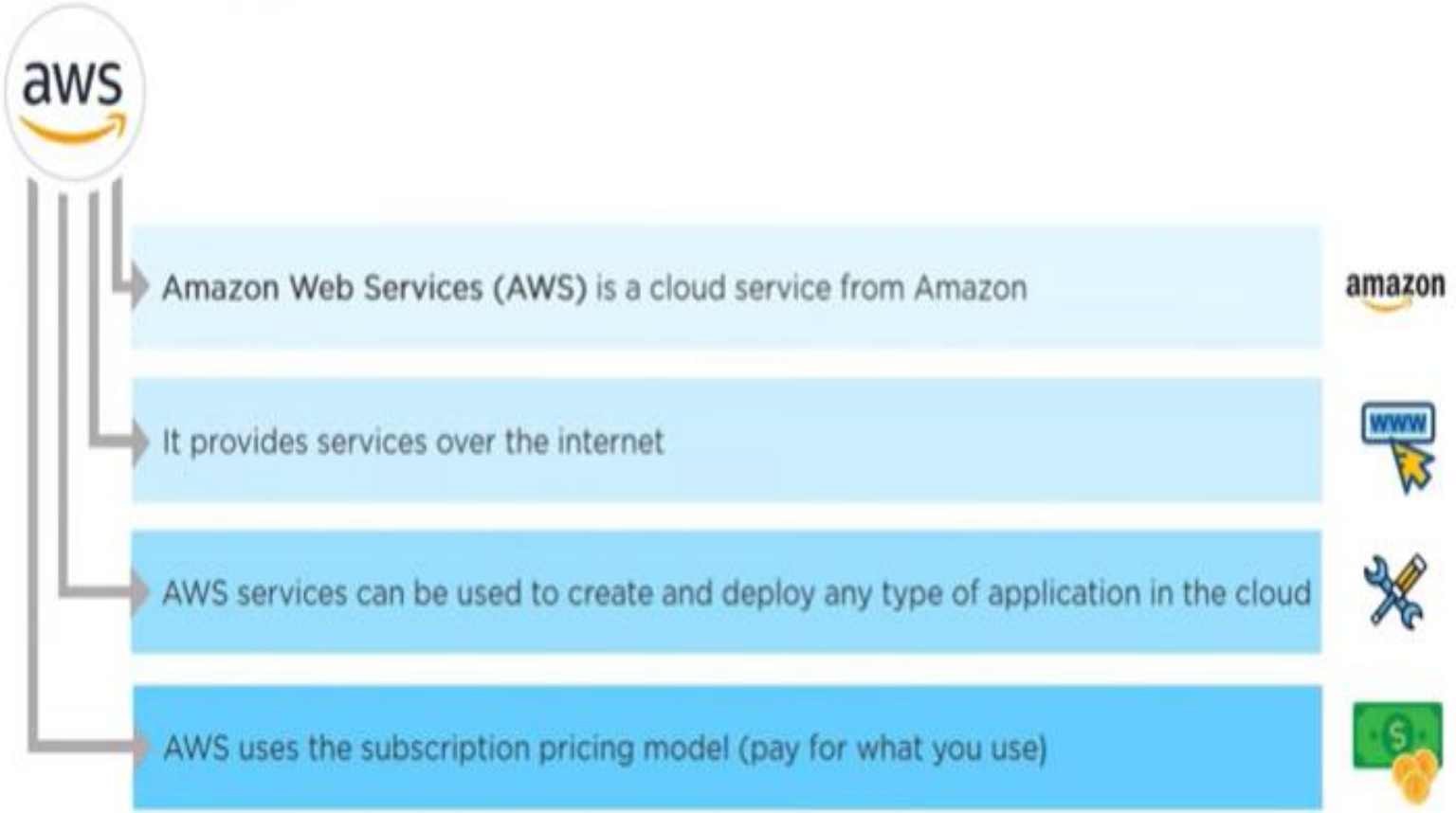
Common Applications in the Cloud

The last section provided a clear outline on the various business benefits of Cloud Computing. However, there exist certain data security and privacy issues while hosting applications on the cloud. Organisations are widely experimenting with different parts of the business that can yield favourable results without falling victim to these concerns.

The most suitable and common applications that can be hosted in the cloud are:

- Development and Testing
- Big Data Analysis
- Websites and CRMs
- Storage and Backup
- Disaster Recovery

Cloud Computing with AWS



AWS Cloud History

- **2004–2006- Launch of AWS**
- AWS officially launched with three core services:
  - **Amazon S3 (Simple Storage Service)**: Object storage for the cloud.
  - **Amazon EC2 (Elastic Compute Cloud)**: On-demand virtual servers.
  - **Amazon SQS (Simple Queue Service)**: Messaging for distributed systems.

- **2007–2010: Early Adoption and Growth**
  - **New Services**:
    - Amazon RDS (Relational Database Service)
    - Amazon CloudFront (Content Delivery Network)

AWS Cloud History

- **2011–2014: Enterprise Adoption and Expansion**
  - **New Innovations**:
    - Amazon DynamoDB: NoSQL database service.
    - Elastic Beanstalk: Platform-as-a-Service for deploying web applications.
    - Redshift: Cloud data warehousing.

- **2015–2018: Leadership and Innovation**
  - **New Innovations**:
    - AWS Lambda: Serverless computing for running code without managing servers.
    - Amazon SageMaker: Machine learning platform for developers.
    - AWS IoT: Internet of Things services.

AWS Cloud History

- **2019–2023: Dominance and Diversification**

  - **New Innovations**:
    - Artificial intelligence and machine learning: Enhanced SageMaker and introduced AI services like Rekognition and Lex.
    - Hybrid Cloud: Launched AWS Outposts for on-premises integration.

AWS Global Infrastructure

- Amazon Web Services (AWS) operates one of the largest and most advanced cloud infrastructures in the world.

- Its global infrastructure is designed to provide highly reliable, scalable, and low-latency cloud services to customers worldwide.

- This is achieved through-
  - **Regions**
  - **Availability Zones**
  - **Edge Locations**.

AWS Global Infrastructure

Regions

•**Definition**: A **Region** is a physical location in the world where AWS has multiple data centers.

•**Features**:

- Each Region is isolated and independent to ensure fault tolerance.
- Consists of multiple **Availability Zones** for redundancy and high availability.

- **Examples**:
  - **US East (N. Virginia)**: us-east-1
  - **Asia Pacific (Mumbai)**: ap-south-1
  - **Europe (Frankfurt)**: eu-central-1

AWS Global Infrastructure

Availability Zones

•**Definition**: AZs are discrete data centers within a Region, each with independent power, cooling, and networking.

•**Features**:

- Typically, each Region has **3 or more AZs** (some may have up to 6).
- Connected via low-latency, high-throughput private fiber links.
- AZs provide fault isolation but allow applications to achieve high availability by spreading across multiple AZs.

•**Benefits**:

- Fault tolerance: Failure in one AZ does not affect others.
- Redundancy for critical workloads.

AWS Global Infrastructure

Edge Locations

•**Definition**: Edge locations are data centers designed to deliver content closer to end users through services like Amazon CloudFront (Content Delivery Network).

•**Features**:
•Over **450+ edge locations** globally (as of 2024).
•Used for caching static content, reducing latency, and improving application performance.
•Support for services like AWS Global Accelerator and Route 53.

Data Center

A **data center** is a facility that houses an organization's critical IT infrastructure, including servers, storage systems, networking equipment, and software applications.

It provides the backbone for modern computing needs, enabling businesses and individuals to store, process, and distribute data efficiently.

Key Components of a Data Center

1. Computing Resources:

•Servers: Physical or virtual machines that run applications, store data, and perform computations.

•High-Performance Computing (HPC): For tasks requiring significant computational power (e.g., AI, scientific simulations).

Data Center

2.Storage Systems:

•Disk Storage: HDDs and SSDs for persistent data storage.

•SAN/NAS: Storage Area Networks (SANs) and Network-Attached Storage (NAS) systems for data sharing and backups.

3.Networking Equipment:

•Routers, switches, and firewalls to manage data traffic within and outside the data center.

•High-speed connectivity for low-latency operations.

4.Power and Cooling Systems:

•Redundant power supplies (UPS, backup generators).

•Cooling systems to maintain optimal temperatures and prevent hardware failures.

Data Center

5.Physical Infrastructure:

- Racks and cabinets for organizing equipment.
- Raised floors or other cable management solutions.

6.Security Systems:

- Physical security: Surveillance cameras, biometric access, and guards.
- Cybersecurity: Firewalls, intrusion detection systems, and encryption.

AWS Data Center

AWS data centers are a critical part of its global infrastructure, ensuring high availability, scalability, and security for its cloud services.

1.Design and Redundancy:

1. AWS data centers are grouped into **Availability Zones (AZs)** within **Regions**.
2. Multiple AZs ensure fault tolerance and disaster recovery capabilities.

2.Security:

1. Physical security measures: 24/7 surveillance, biometrics, and perimeter fencing.
2. Cybersecurity: Data encryption, strict access controls, and compliance with global standards (e.g., GDPR, HIPAA).

AWS Data Center

3. **Energy Efficiency**:
    1. AWS is committed to sustainability and uses renewable energy sources to power its data centers.
    2. Advanced cooling systems to reduce energy consumption.

4. **Scalability**:
    1. AWS data centers support rapid scaling of services, accommodating fluctuating customer demands.

Functions of Data Center

1. Data Storage and Backup:

•Ensures critical data is stored securely and can be recovered if needed.

2. Hosting Applications:

•Runs business applications, websites, and databases.

3. Data Processing:

•Handles computational tasks like analytics, machine learning, and simulations.

4.Disaster Recovery:

•Provides redundancy and ensures business continuity in case of system failures.